

WHITMAN 5 ENE, NE, USA

WMO: 744360

Lat: 42.0680N

Lon: 101.4450W

Elev: 3740

StdP: 12.82

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 94079

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -5.3	(c) -0.5	(d) -11.0	(e) 3.5	(f) -4.3	(g) -6.5	(h) 4.5	(i) 0.1	(j) 33.2	(k) 13.8	(l) 30.4	(m) 16.6	(n) 9.7	(o) N/A	(p) 0.641

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 23.1	(c) 93.3	(d) 65.2	(e) 89.8	(f) 65.5	(g) 86.7	(h) 64.9	(i) 70.9	(j) 82.9	(k) 69.4	(l) 81.4	(m) 68.1	(n) 80.4	(o) 11.6	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 67.7	(b) 117.2	(c) 75.0	(d) 65.9	(e) 109.9	(f) 74.4	(g) 64.3	(h) 103.9	(i) 73.1	(j) 37.3	(k) 83.1	(l) 36.0	(m) 81.5	(n) 34.7	(o) 80.5	(p) 75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.5	25.4	22.2	-12.4	98.6	3.5	4.0	-14.9	101.5	-16.9	103.8	-18.9	106.0	-21.4	108.9
			-12.1	73.1	3.5	1.5	-14.6	74.1	-16.7	75.0	-18.7	75.8	-21.2	76.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.3	26.4	26.4	38.3	45.6	55.4	67.0	73.1	70.6	62.7	48.5	37.9	26.3		
	DBStd	19.56	12.07	13.03	11.81	10.36	9.44	6.78	5.78	5.85	8.43	10.08	11.46	12.20		
	HDD50	3317	734	664	390	198	55	0	0	0	9	148	385	734		
	HDD65	6757	1198	1081	828	585	317	54	8	20	138	515	814	1199		
	CDD50	2700	1	3	27	65	224	511	717	639	391	101	21	0		
	CDD65	666	0	0	0	2	21	115	260	194	70	4	0	0		
	CDH74	7116	0	0	14	58	309	1238	2691	1864	846	90	6	0		
	CDH80	2806	0	0	0	11	80	439	1198	746	313	19	0	0		
(14) Wind	WSAvg	10.7	10.7	11.0	11.5	13.0	11.8	10.6	9.5	9.1	9.9	10.8	10.7	10.4		
(15) Precipitation	PrecAvg	20.7	0.4	0.5	1.1	2.4	2.9	3.9	2.9	2.2	1.7	1.7	0.7	0.4		
	PrecMax	27.3	0.9	1.2	1.9	4.4	6.3	5.8	7.0	3.7	3.5	3.9	2.0	0.7		
	PrecMin	11.7	0.0	0.0	0.2	0.3	0.9	0.6	0.7	0.9	0.6	0.0	0.0	0.0		
	PrecStd	3.7	0.2	0.3	0.5	1.1	1.4	1.4	1.5	0.8	0.9	1.0	0.5	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.9	63.9	76.0	81.4	87.2	94.8	99.7	95.9	93.1	82.6	74.5	58.1		
		MCWB	42.5	45.0	50.4	52.5	60.8	64.9	65.7	64.5	62.5	55.7	50.0	42.4		
	2%	DB	52.5	56.0	68.5	74.0	82.0	89.4	94.2	91.4	87.9	75.8	65.5	51.4		
		MCWB	40.3	41.8	47.3	52.7	59.4	65.6	65.9	64.7	62.3	53.9	46.5	39.5		
	5%	DB	46.9	49.7	62.8	68.6	77.1	85.0	90.5	87.6	83.5	70.3	60.0	46.9		
		MCWB	37.2	38.4	45.3	50.0	58.1	64.8	67.0	64.6	61.9	52.0	44.3	36.8		
	10%	DB	42.5	44.2	56.9	63.1	72.4	81.1	87.0	84.2	78.6	65.0	54.8	42.4		
		MCWB	35.1	35.7	42.5	47.8	55.7	63.7	66.8	64.8	59.6	49.8	42.3	34.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.5	46.6	53.4	56.0	64.7	71.4	73.0	71.5	69.0	59.1	51.3	44.1		
		MCDB	56.0	63.1	69.7	71.8	77.4	83.1	85.9	84.2	83.4	74.7	66.1	54.3		
	2%	WB	39.8	42.4	49.5	53.6	62.6	69.4	71.4	69.8	66.4	56.1	47.4	40.5		
		MCDB	49.9	56.0	65.8	68.4	75.6	80.6	83.2	81.4	80.5	69.8	63.1	51.1		
	5%	WB	37.3	39.4	46.9	51.3	60.0	67.6	70.2	68.4	64.4	53.6	44.5	37.3		
		MCDB	44.9	48.8	62.3	66.5	71.5	80.5	82.3	79.7	77.5	65.7	58.5	46.1		
	10%	WB	35.0	36.5	44.3	49.0	57.6	65.8	68.6	67.1	62.5	51.2	42.1	34.8		
		MCDB	41.9	44.7	57.7	61.2	68.5	77.9	81.2	78.0	74.0	63.7	53.3	42.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.7	19.2	21.8	21.6	21.3	22.2	23.1	22.7	23.7	21.8	21.1	17.6		
		MCDBR	24.7	25.7	30.8	31.0	28.5	27.7	28.1	28.3	30.3	30.5	28.4	22.5		
		MCWBR	14.3	14.8	15.0	13.8	11.9	10.6	8.6	9.5	10.7	13.8	13.9	13.9		
	5% WB	MCDBR	22.1	26.1	30.3	28.3	25.1	25.0	24.2	23.4	26.0	27.8	27.0	22.5		
		MCWBR	14.1	15.7	15.4	13.8	11.5	11.8	10.0	9.3	11.5	13.6	13.6	14.4		
(40) Clear-Sky Solar Irradiance	taub		0.251	0.259	0.282	0.312	0.334	0.351	0.373	0.367	0.328	0.289	0.261	0.253		
	taud		2.556	2.533	2.504	2.423	2.443	2.422	2.400	2.398	2.485	2.562	2.562	2.531		
	Ebn at Noon		294	307	309	304	298	292	284	283	288	291	286	282		
	Edh at Noon		21	25	30	35	35	36	37	36	30	25	21	20		
(44) All-Sky Solar Radiation	RadAvg		711	996	1356	1676	1970	2255	2251	1936	1569	1090	773	605		
	RadStd		49	52	101	90	143	146	73	84	75	98	44	37		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page